19. VSEPR theory is used to predict the shape and bond angle of molecules.
    a) Write the postulates of VSEPR theory. (2)
    b) Explain the shape and bond angle of NH_3 molecule using VSEPR theory. (2)
    c) PCl_5 molecule is unsymmetric. Why? (2) *[September 2016]*
20. The geometry of the molecule is decided by the type of hybridisation.
    a) Discuss the shape of PCl_5 molecule using hybridisation. (2)
    b) Give the reason for the high reactivity of PCl_5. (2)
    c) Isoelectronic species have the same bond order. Among the following choose the pair having same bond order.
    CN^-, O_2^-, NO^+, CN^+ (1) *[March 2017]*
21. a) The hybridization of C in ethene is ....
    i) sp ii) sp^2 iii) sp^3 iv) sp^3d (1)
    b) Explain sp^3d^2 hybridization with an example. (3)
    c) Calculate the bond order of Lithium molecule. (At. no. of Li is 3) (1) *[July 2017]*
22. Predict the shape of XeF_4 molecule, according to VSEPR theory. (1)
23. By using the concept of hybridization, explain the structure of H_2O molecule. (2)
24. Write the molecular orbital electronic configurations of N_2 and O_2 and calculate their bond orders. Give a comparison of their stability and magnetic behaviour. (4) *[March 2018]*
25. If Z-axis is the internuclear axis, name the type of covalent bond formed by the overlapping of two p_y-orbitals. (1)
26. Write any two limitations of octet rule. (2)
27. The diatomic species Ne_2, does not exist, but Ne_2^- can exist. Explain on the basis of molecular orbital theory. (4) *[August 2018]*
28. Represent the Lewis structure of Ozone (O_3) molecule and assign the formal charge on each atom. (2)
29. Among NaCl, $BeCl_2$ and $AlCl_3$, which one is more covalent? Justify the answer. (2)
30. Write the molecular orbital electronic configuration of N_2 and O_2 molecules. Compare the stability and magnetic behaviour of these molecules on the basis of M. O. theory. (3) *[March 2019]*
31. The dipole moment of BeF_2 is zero, while that of H_2O is 1.85 D. Account for this the on basis of their molecular structure. (2)
32. (a) A molecule of the type AB_4E has 4 bond pairs of electrons and 1 lone pair ofelectron. Predict the most stable structure of this compound. (1)
    (b) Hydrogen fluoride is a liquid, while hydrogen chloride is a gas. Why ? (1)
33. Draw the molecular orbital diagram for F_2 molecule. Account for its magnetic character. (3) *[July 2019]*
34. (a) Give two examples of compounds having expanded octet. (1)
    (b) Draw the Lewis dot symbols of (i) Cl_2 (ii) NF_3 (2)
35. (a) Predict the hybridisation of phosphorous atom in PCl_5 molecule. (1)
    (b) Account for the high reactivity of PCl_5 molecule. (1)
    (c) Draw the MO energy level diagram of O_2 molecule. (2) *[March 2020]*
36. (a) Define Bond angle. (1)
    (b) NH_3 and NF_3 molecules have a pyramidal shape with a lone pairs of electrons on nitrogen atom. But the dipole moment of NH_3 is 4.9×10^{-30} Cm and that of NF_3 is 0.8×10^{-30} Cm. Give reason. (2)
37. (a) The bond angle in water is lower than the tetrahedral angle. Why? (1)
    (b) Give 1 example of a molecule in which the central atom is in sp hybridisation. Predict its geometry. (1)